SQL Server Operations

1. Backup strategy

Full backups establish a baseline. Differential backups capture changes since the latest full backup. Transaction log backups support point-in-time recovery under FULL recovery model.

Practical guidance

Example schedule: full backup Sunday 02:00, differential daily 02:00, log backup every 15 minutes.

Training reference: synthetic example content for a multi-document RAG assignment. Not an operational policy.

SQL Server Operations

2. TempDB management

TempDB supports temporary objects, sorts, hash operations, and version storage. Multiple equally sized data files may reduce allocation contention.

Practical guidance

Avoid routine shrinking. Diagnose growth, active sessions, and configured file sizes before considering a restart.

Training reference: synthetic example content for a multi-document RAG assignment. Not an operational policy.

SQL Server Operations

3. Index maintenance

Use execution plans, index usage, and fragmentation alongside workload evidence. Rebuilds can consume substantial log space.

Practical guidance

Evaluate statistics updates separately from rebuilds; schedule disruptive work in maintenance windows.

Training reference: synthetic example content for a multi-document RAG assignment. Not an operational policy.

SQL Server Operations

4. Monitoring and alerts

Monitor CPU, memory grants, waits, blocking, deadlocks, disk latency, and backup success. Alerts should identify affected instance and time window.

Practical guidance

Investigate recurring CPU alerts with Query Store and execution plans rather than treating CPU alone as root cause.

Training reference: synthetic example content for a multi-document RAG assignment. Not an operational policy.

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5. Recovery testing

A backup is not proven until restored. Perform periodic restore drills and DBCC CHECKDB checks on restored copies.

Practical guidance

Target RPO: 15 minutes. Target RTO: 2 hours. These are illustrative policy values for this sample handbook.

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